Micropin Rev. 2 hardware findings and MAME back-port audit

What is folded in, what was checkpointed, and what remains provisional - 4 September 2026

Bottom line

Most discoveries are represented in the shared Rev. 2 driver or development layout, and the latest bridge, tester, cabinet-input and lamp-layout work is now checkpointed. They are not all finished: several input-timing behaviors are pragmatic simulations rather than established hardware models, and the stock pentacup2 entry still uses the old layout and stale not-working/no-sound declarations.

Repository state

Item	State
MAME branch	codex/pencupx-experimental at eb0eaa7e (Complete Micropin arbiter protocol and hardware tester).
Checkpoint	Focused micropindev build and MAME validation for pencupx/pencuptst passed before commit.
Included	Latest arbiter protocol, standalone tester, corrected rollover lamp assignments, lamp 0/6 additions and cabinet Start routing.
Stock user target	pentacup2 still selects the historical layout and advertises MACHINE_NO_SOUND / MACHINE_NOT_WORKING. pencup2d is the practical four-ROM development target.

Status legend

FOLDED = committed shared driver or layout contains it. PROVISIONAL = useful behavior, but not a faithful hardware conclusion. OPEN = still needs reverse engineering or hardware validation.

Core, memory and interrupts

Finding	Confidence	MAME status	Back-port/action
Rev. 2 uses an 8085 modeled at 1.5 MHz (believed 3 MHz crystal divided by two).	Medium	FOLDED	Confirm with scope/crystal marking when convenient.
Base map: ROM $0000-$1fff; backed work RAM $2000-$23ff; display writes $23c0-$23df.	High	FOLDED	Retain.
Fifth 2 KiB ROM is decoded at $2800-$2fff. Experimental bidirectional expansion aperture is $3000-$31ff.	High for current board plan	FOLDED	Retain separate pencup2c and pencupx maps until physical decode is verified.
MAME drives alternating RST 5.5/6.5 phases from a 454 Hz callback, giving a 227 Hz edge cadence per line.	Medium-low	FOLDED / PROVISIONAL	Measure the real interrupt signals. The existing cadence came from the established emulator model.
All work RAM is battery-backed and can be poisoned by interrupted writes. Boot while holding right flipper invokes hard recovery at $005d.	High	FOLDED	Forced clear/recovery is intentionally applied on every normal Rev. 2 MAME boot, with a warning that high scores are lost.

Inputs: the important architectural correction

The 32 playfield contacts are not a conventional keyboard-style switch matrix. The inductive sensing hardware continuously presents per-contact state in RAM at $23e0-$23ff. The old row/column names are compatibility labels inherited from PinMAME, not the motherboard programming model.

Finding	MAME status	Remaining concern
32 inductive samples at $23e0-$23ff; game dispatch table at $141f-$145e.	FOLDED	MAME writes inferred contact bytes directly. It does not model the physical ring-time measurement/DMA circuitry.
Reference/adjusted bank near $233b-$235a participates in contact qualification.	OPEN	Exact ownership and comparison semantics remain incompletely understood.
The apparent 'same rollover cannot score twice until another switch' behavior.	PROVISIONAL	Current interrupt suppression/pulse behavior makes play practical, but the actual hardware/software latch mechanism remains unresolved.
Outhole behaves like a toggled/flip-flop contact in the proven PinMAME model.	FOLDED / PROVISIONAL	Keep until a logic trace proves a more faithful representation.

Playfield contact mappings

Group	Established mapping	MAME status
Core tests	R-K inlane = contact 1; 25-point bumper Y-A = contact 15; W-K 10,000/extra-ball = contact 16; outhole R-A = contact 25; Q-A inlane = contact 32.	FOLDED and gameplay-tested.
Cups 1-5	R-H, R-F, R-S, E-G, E-D -> contacts 30, 28, 26, 21, 19.	FOLDED; pairing was gameplay-tested.
Standups 1-5	R-G, R-D, E-H, E-J, E-K -> contacts 29, 27, 22, 23, 24. Third/fifth physical pairing was corrected during play tests.	FOLDED.
Eight 500-point rollovers	E-F, W-S, W-D, W-G, W-H, W-F, E-A, E-S -> contacts 20, 10, 11, 13, 14, 12, 17, 18.	FOLDED.
Shared standup bar	ROM handler $0784 sets CONTROL_FLAGS bit $20; target code can then fire coil output 18.	PROVISIONAL shortcut: MAME's H key directly sets RAM bit $2191. Proper port-1 stimulus should replace this.

Cabinet switch fan-out

Physical observation says there is one contact per flipper button, but each contact serves two software paths. The working emulator model fans each button into the auxiliary service path and port 4.

Control	Current working model	Status
Left flipper	Auxiliary bit $10 plus port 4 bit $40.	FOLDED, but exact physical fan-out still deserves ROMulator verification.
Right flipper	Auxiliary bit $20 plus port 4 bit $20. Synthetic Enter launch also temporarily asserts this path.	FOLDED. Port 4 bit $20 is corroborated by boot recovery.
Enter Players / Start	Presented through auxiliary bit $40; it must not leak as port 4 bit $80.	FOLDED in eb0eaa7e. Real arb_tester behavior exposed the earlier mistake.
Credit	Port 4 bit $10 in the current model.	Working; retain pending hardware trace.

Caution

Experiments that simply swapped left/right port-4 masks broke unrelated switch behavior and were reverted. Treat the current mapping as the best working model, not final proof of every trace on the motherboard.

Outputs

Subsystem	Learned programming model	MAME status / action
Lamps	Eight active-low bytes on output ports $00-$04 and $0d-$0f: 64 possible outputs, many unused. The game ROM's two lamp images implement flashing in software.	Direct ports are FOLDED. Remove MAME's semantic two-page/2 Hz flash reconstruction and expose literal time-ordered writes; this TODO is already in source.
Score-panel LEDs	Additional indicators are multiplexed through selected display bytes rather than the eight ordinary lamp ports.	FOLDED as extra lamp outputs. Names/physical labels can be refined.
Coils	Four active-low bytes on ports $05-$08. Game shadows are $21c2-$21c5, then complemented and written to ports. Output N = port $05 + floor(N/8), bit N mod 8.	FOLDED. Full 0-31 mechanism table is documented in symbols.yml.
Sound	Port $0a carries complemented pitch. Port $09 controls enable/duration; stock playback resets/arms with $ff before pitch/duration. Sequence terminator $ff is not a tone.	FOLDED and heard on real hardware. MAME pitch/duration conversion remains empirical pending oscillator schematic.
Displays	Writes to $23c0-$23df feed paired BCD/7448-style digits. Values A-E yield the observed odd glyphs; F is blank. High score uses $23d3-$23d5.	FOLDED in code; complete useful presentation is currently in development layout.

Confirmed coil examples

Launcher output 5; lower bumpers 10/11; right and left flippers 12/13; upper bumper 16; center rollover mechanism 17; standup bar 18; slingshots 19/20; cup 2 output 21; side cup 25; mechanical coin counter 27. Output 18 is also corroborated statically by ROM $0da2-$0db0.

Lamp artwork status

The development layout contains the corrected rollover output assignments, the back-panel lamp 0, an approximate playfield position for otherwise-missing lamp 6, the cup/target mappings, a numbered lamp matrix, high score, bonus and ball state. The newest rollover corrections and lamp 0/6 additions are committed in eb0eaa7e.

MAME consolidation checklist

Priority	Task	Why
Done	Build, validate and commit the latest driver/layout/tester checkpoint as eb0eaa7e.	Protects the protocol, tester, Start routing and lamp corrections.
1	Replace semantic lamp-page reconstruction with literal latch writes (plus visual persistence only if necessary).	The pages and flashing are ROM software behavior, not separate lamp hardware.
2	Route the shared standup-bar test through the correct input/interrupt path instead of directly setting $2191 bit $20.	Current shortcut proves game behavior but bypasses the hardware contract.
3	Validate cabinet fan-out and Start/Credit bits using the ROMulator, then simplify comments and input code around confirmed facts.	Current model works, but contradictory early observations were never resolved electrically.
4	Capture real interrupt cadence and inductive DMA behavior.	Removes reliance on PinMAME timing and the current RST suppression workaround.
5	Promote the Rev. 2 layout/sound/input work from pencup2d into pentacup2 and update machine flags.	Ordinary MAME users should receive the completed work without a development-only system name.
6	Update the stale top-of-file TODO block and add concise hardware notes/tables in-tree.	The source still says Rev. 2 outputs, inputs, displays and sound are unavailable.
7	After physical verification, upstream the driver in reviewable slices: core/map, displays, outputs/sound, inputs, artwork, bridge kept experimental.	Separates broadly useful emulation from the Micropin-specific development transport.

Already shared across variants

The C++ implementations for direct outputs, sound, display decoding, NVRAM recovery, switch helpers and save-state registration live in pent8085_state and are inherited by Rev. 2 variants. Variant differences are chiefly memory maps, ROM filenames, bridge/test flags and selected layout. Thus most back-port work is consolidation and fidelity cleanup, not reimplementation.

Source landmarks

MAME driver: /Users/msw/dev/micropin/upstream/mame/src/mame/pinball/micropin.cpp
Development layout: /Users/msw/dev/micropin/upstream/mame/src/mame/layout/micropin2dev.lay
Hardware symbols/tables: /Users/msw/dev/micropin/symbols.yml
Arbiter: /Users/msw/dev/micropin/arb/arb.asm
Python bridge: /Users/msw/dev/micropin/romulator/micropin_bridge.py